Year 11 Foundation

REVISION PACK

Algebra 7

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Algebra 7 · Formulae (Lessons 2.1, 2.2)

Q	Working	Answer
substitution into P = 2l + 2w		
1.1	$P = 2 \times 8 + 2 \times 5$ $= 16 + 10$ $= 26$	$P = 26$
1.2	$P = 2 \times 12 + 2 \times 7$ $= 24 + 14$ $= 38$	$P = 38$
1.3	$P = 2 \times 3.5 + 2 \times 2$ $= 7 + 4$ $= 11$	$P = 11$
1.4	$P = 2 \times 10 + 2 \times 4.5$ $= 20 + 9$ $= 29$	$P = 29$
1.5	$P = 2 \times 15 + 2 \times 9$ $= 30 + 18$ $= 48$	$P = 48$
1.6	$P = 2 \times 6 + 2 \times 2.5$ $= 12 + 5$ $= 17$	$P = 17$
substitution into v = u + at		
1.7	$v = 10 + 4 \times 3$ $= 10 + 12$ $= 22$	$v = 22$
1.8	$v = 5 + 6 \times 7$ $= 5 + 42$ $= 47$	$v = 47$

1.9	$v = 15 + 2 \times 9$ $= 15 + 18$ $= 33$	$v = 33$
1.10	$v = 3 + 5 \times 8$ $= 3 + 40$ $= 43$	$v = 43$
1.11	$v = 20 + 3 \times 4$ $= 20 + 12$ $= 32$	$v = 32$
1.12	$v = 7 + 9 \times 2$ $= 7 + 18$ $= 25$	$v = 25$
using a formula		
1.13	$35n + 15 = 155$ $35n = 140$ $n = 4$	**4 hours**
1.14	$25n + 10 = 135$ $25n = 125$ $n = 5$	**5 hours**
1.15	$50n + 30 = 280$ $50n = 250$ $n = 5$	**5 hours**
1.16	$45n + 20 = 200$ $45n = 180$ $n = 4$	**4 hours**
1.17	$60n + 40 = 340$ $60n = 300$ $n = 5$	**5 hours**
1.18	$20n + 15 = 135$ $20n = 120$ $n = 6$	**6 hours**

rearranging formulae		
1.19	**(a)** $y-2=5x$ $x=\dfrac{y-2}{5}$ **(b)** $w=\dfrac{A}{l}$	**(a)** $x=\dfrac{y-2}{5}$ **(b)** $w=\dfrac{A}{l}$
1.20	**(a)** $y+7=2x$ $x=\dfrac{y+7}{2}$ **(b)** $h=\dfrac{V}{A}$	**(a)** $x=\dfrac{y+7}{2}$ **(b)** $h=\dfrac{V}{A}$
1.21	**(a)** $y-1=6x$ $x=\dfrac{y-1}{6}$ **(b)** $d=\dfrac{C}{\pi}$	**(a)** $x=\dfrac{y-1}{6}$ **(b)** $d=\dfrac{C}{\pi}$
1.22	**(a)** $y+4=3x$ $x=\dfrac{y+4}{3}$ **(b)** $s=\dfrac{P}{4}$	**(a)** $x=\dfrac{y+4}{3}$ **(b)** $s=\dfrac{P}{4}$
1.23	**(a)** $y-3=8x$ $x=\dfrac{y-3}{8}$ **(b)** $a=\dfrac{F}{m}$	**(a)** $x=\dfrac{y-3}{8}$ **(b)** $a=\dfrac{F}{m}$
1.24	**(a)** $y+9=4x$ $x=\dfrac{y+9}{4}$ **(b)** $t=\dfrac{D}{s}$	**(a)** $x=\dfrac{y+9}{4}$ **(b)** $t=\dfrac{D}{s}$